



1. Vanishing Gradient Problem

This is the **biggest issue**.

When training an RNN using backpropagation through time, gradients get multiplied repeatedly.

- ✦ If values $< 1 \rightarrow$ gradients shrink $\rightarrow$ become almost zero
- ✦ Early layers (long-term dependencies) **stop learning**

Result:

- Model **forgets long-term information**
- Can't learn relationships like:
-
- "I grew up in France... I speak ____" $\rightarrow$ should be French

2. Exploding Gradient Problem

Opposite of vanishing.

- ✦ If values $> 1 \rightarrow$ gradients grow exponentially
- ✦ Weights become huge $\rightarrow$ training becomes unstable

Result:

- Model outputs become unpredictable
- Requires tricks like gradient clipping

3. Short-Term Memory Only

RNNs are **biased toward recent inputs**.

- ✦ Information from earlier timesteps fades away quickly

Example:

Sentence:

"The movie that I watched last year was really amazing"

RNN struggles to connect:

- "movie" $\rightarrow$ "amazing" (long gap)

4. No Explicit Memory Control

Everything is mixed together:

- No mechanism to decide:
- what to **remember**
- what to **forget**
- what to **output**

Result:

- Important info gets overwritten
- Irrelevant info is kept

5. Difficulty Learning Long-Term Dependencies

Even beyond gradients:

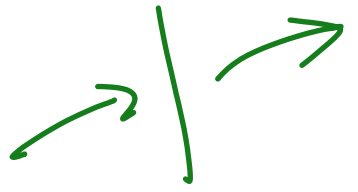
RNN has no structured way to store long-term context

Result:

- Performs poorly on:
- language translation
- long sentences
- time-series with long gaps

simple

Hi this is Akam ;
[learning became deep]



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